Switch Concept & Architecture

What types of architecture makes sense ?

How to reduce the effect of blocking ?

Example architectures

Switching Principles

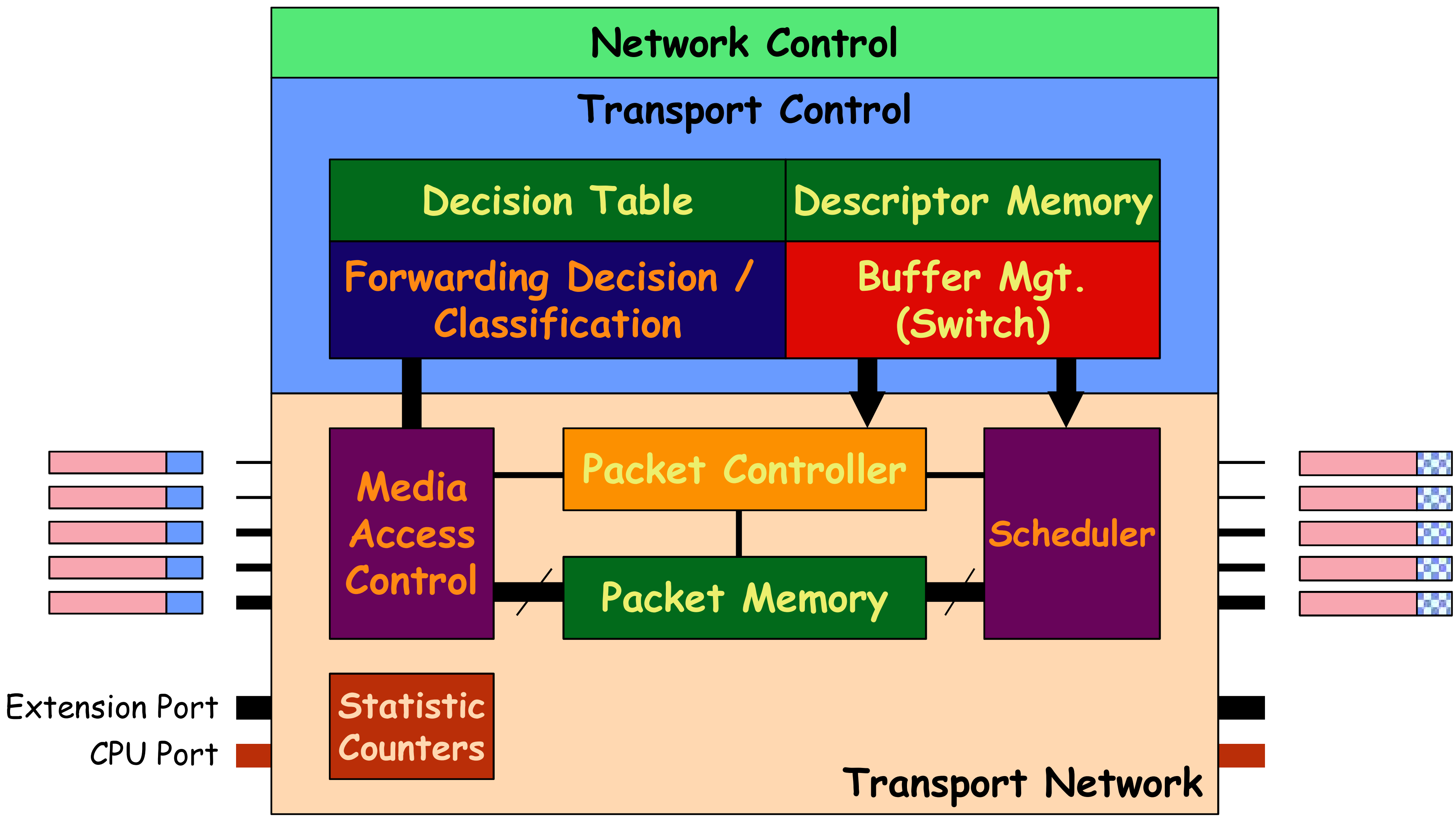
- Switching is the transport of information/data from an *incoming logical channel* to an *outgoing logical channel*.
- Logical channels are characterized by
  - » physical inlet / outlet identified by a physical port number
  - » logical channel on the physical port identified by
    - ATM: VPI / VPI
    - L2: DA / SA
    - L3: IP Address
    - Multi-layer: Flow, TCP port, etc.
- Switch operations
  - » cells / packets arrive at the input port
  - » header lookup, switching decision and translation is performed
  - » packets are routed through or queued in the switch to the appropriate output port.

Switching Principles

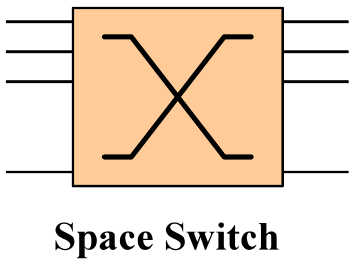
Switches have three main part.

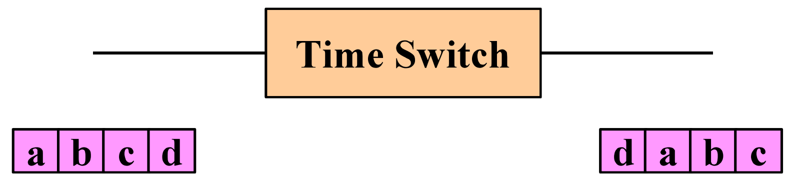
- Transport Network:
  - » Physical means for switching cells from one input port to one (or more) output port(s) in the switch
  - » Performs functions of the User Plane
- Transport Control
  - » Controls the transport network based on analysis of signalling information
    - decides which inlet connects to which outlet via routing, switching, etc. ...
- Network Control
  - » Sets *transport control* parameters / tables.
    - Routing table, resource mgt, call admission control (CAC), etc.
  - » Performs functions in the Control Plane
    - Routing, network mgt., signalling, signalling AAL (SAAL), etc.

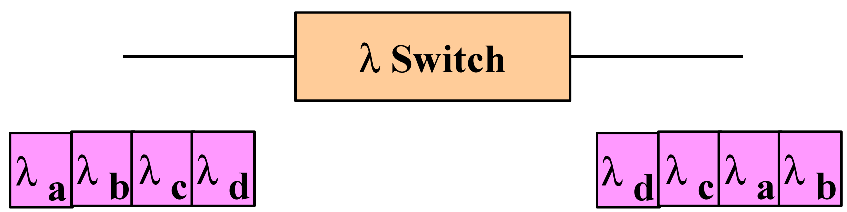
Switching Components



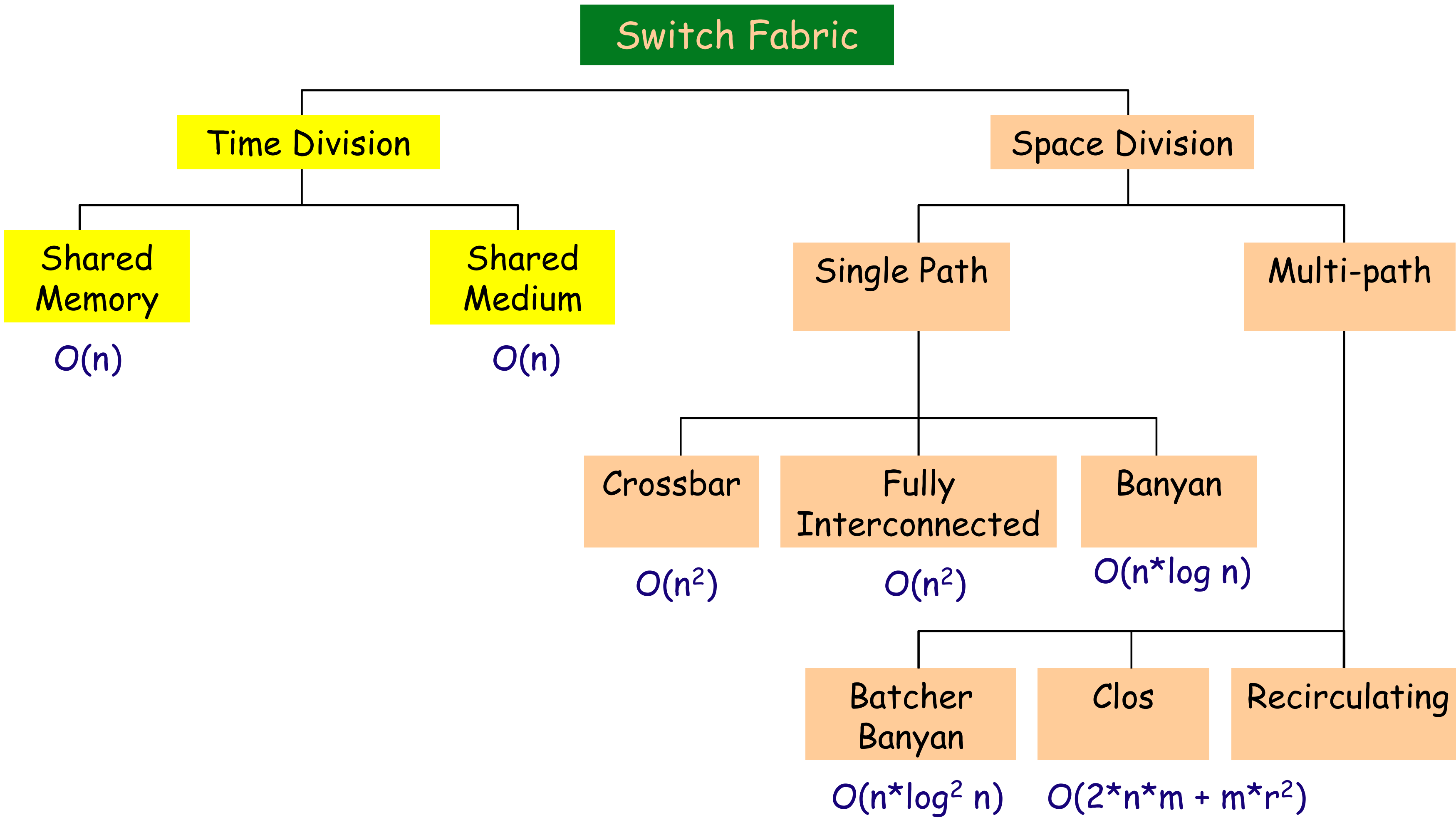
Classical Switching Systems



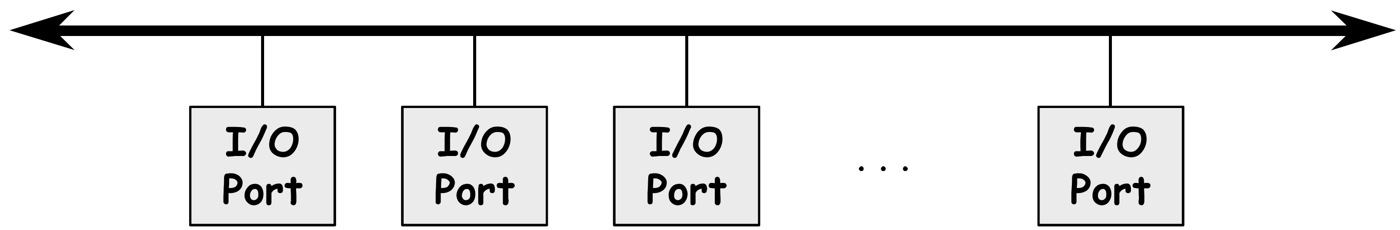




Switching Fabric Classification

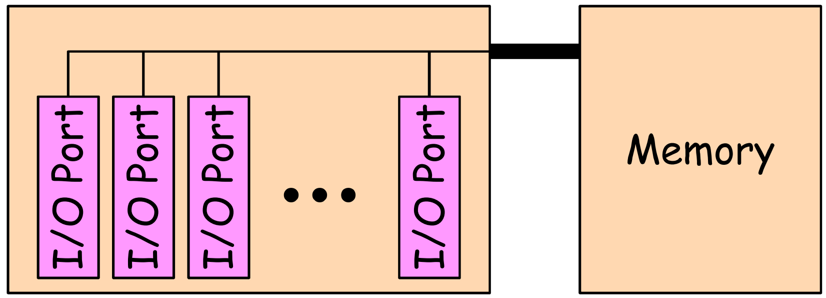


Backplane Switching System



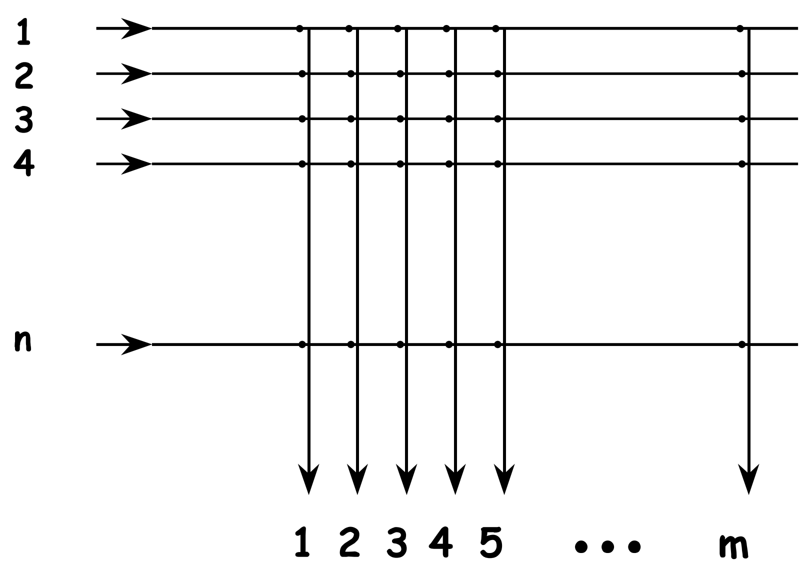
- The Bus is used to connect the inlet / outlet ports together.
  - » Cells are transported via the *Bus*.
  - » Outlet ports accepts cells/packets based on their address
- Non-blocking bus: Σ [Port speed] < Bus throughput rate
  - » Blocking bus effect can be reduced via statistical muxing effect.
- Advantages:
  - » Multicasting is easily supported
  - » Bandwidth limited to about 2 Gbps.
  - » Easy integration with existing LAN equipment based on backplane technology, such as hubs.

Shared Memory Switching System



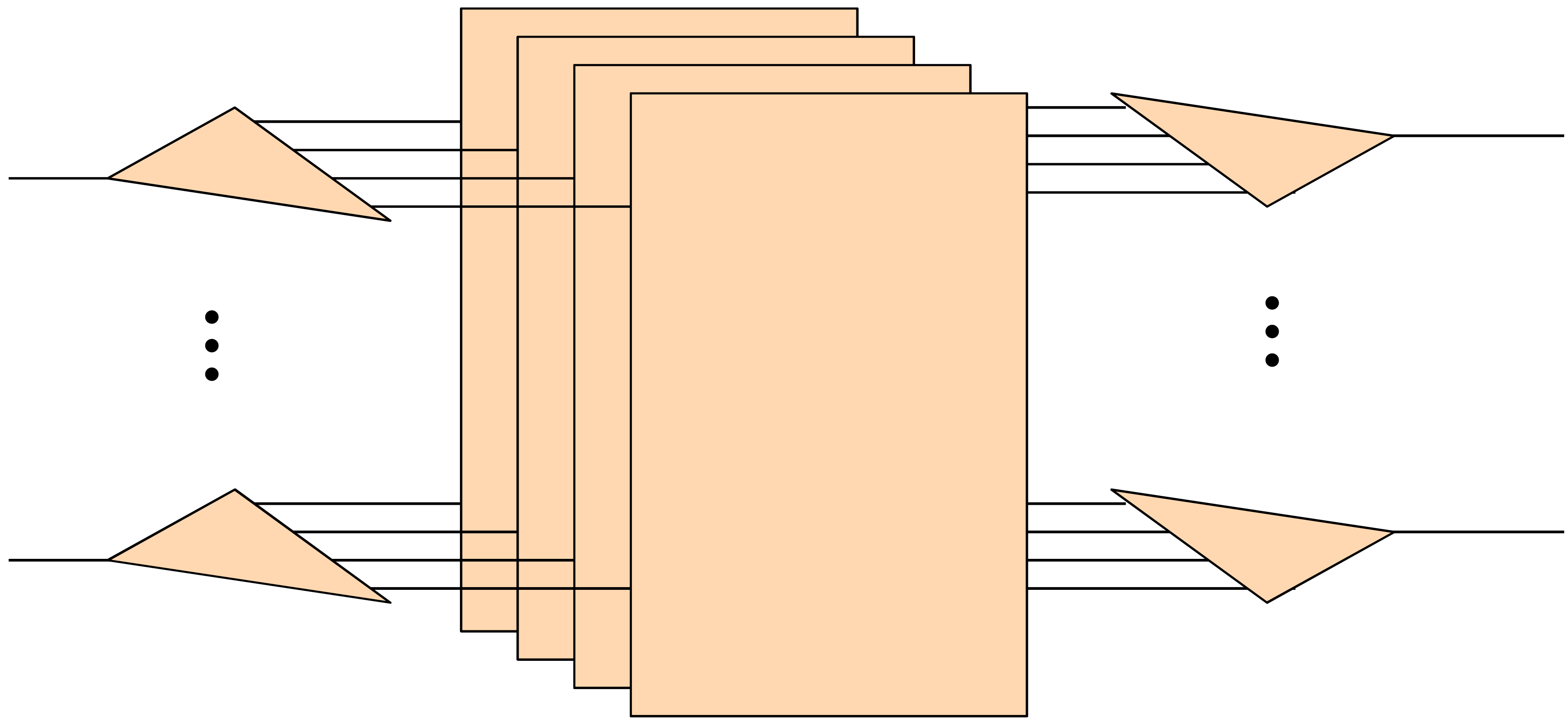
- The Shared Memory is used to connect the inlet / outlet ports together.
  - » Cells are temporarily buffered
  - » Outlet ports pulls cells/packets based on their queue
- Non-blocking memory bandwidth required:
  - ≈ Σ [Port speed] < Memory throughput rate
- Advantages:
  - » Memory usage reduced via sharing
  - » Multicasting is easily supported
  - » Bandwidth limited by memory technology.

Crossbar Switching System

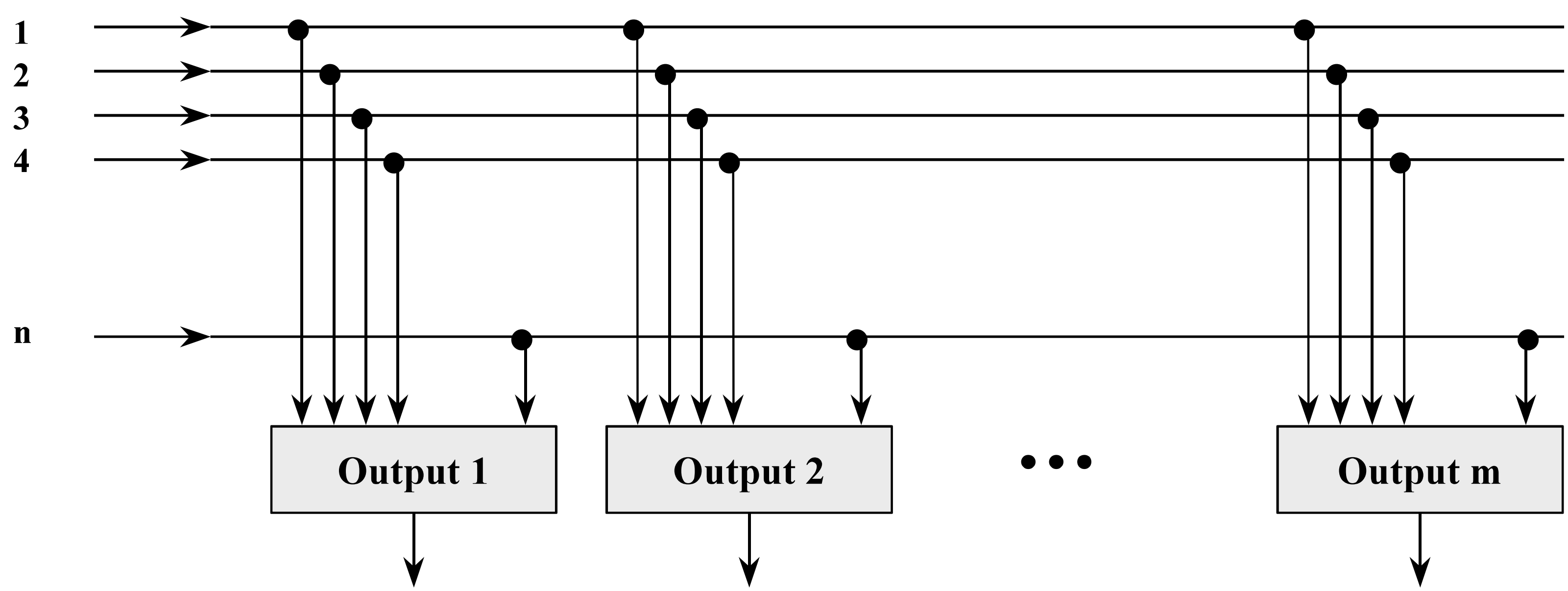


- Internally non-blocking:
  - » each I/O pair has an unique path.
  - » externally blocking at the output.
- Advantage:
  - » high speed internal paths
  - » multicast ready
  - » simple
- Disadvantage:
  - » difficult to scale
  - » input queueing
  - » switching elements, $O(n^2)$

Multiplane Network

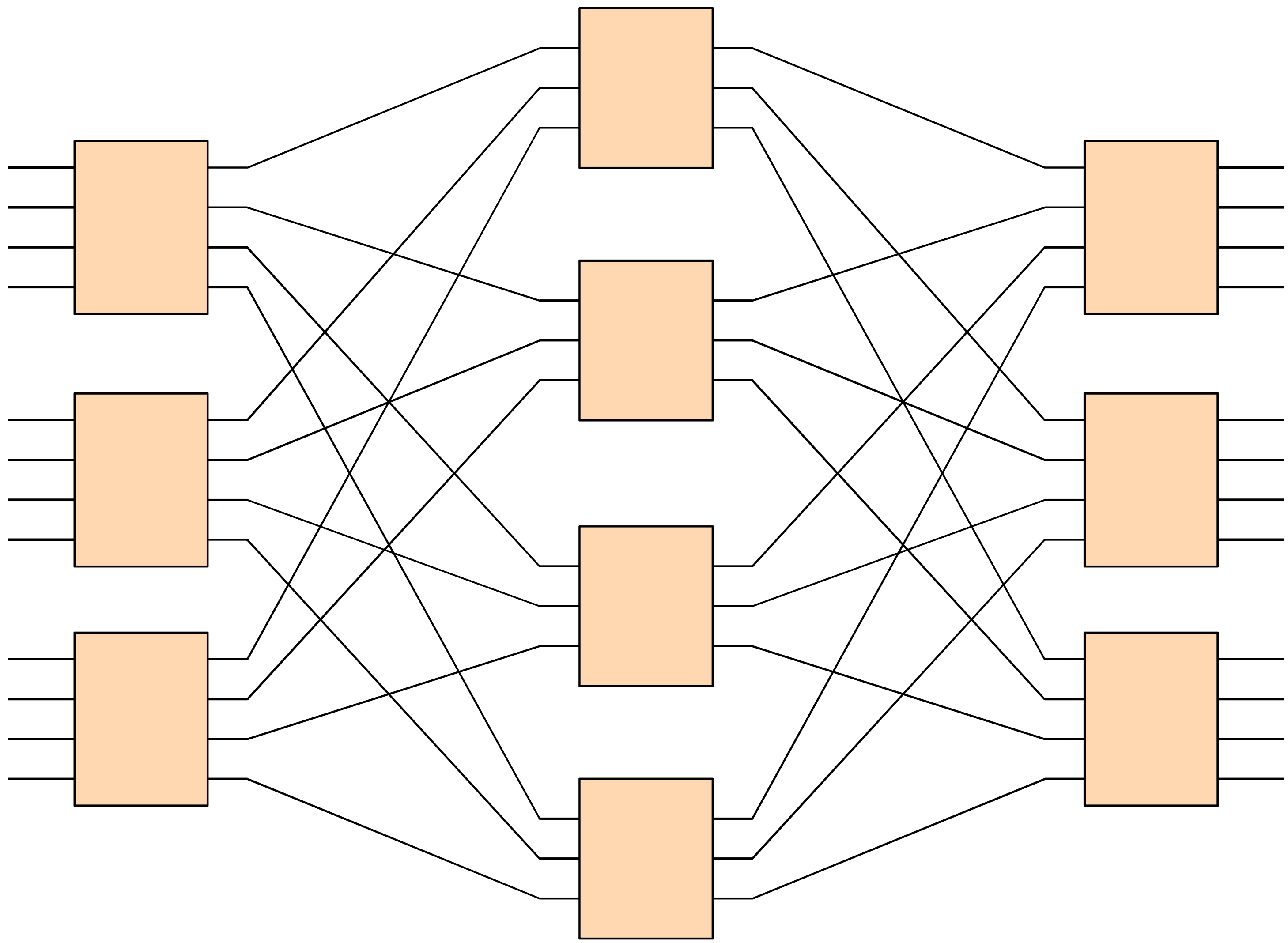


Knockout Switching System

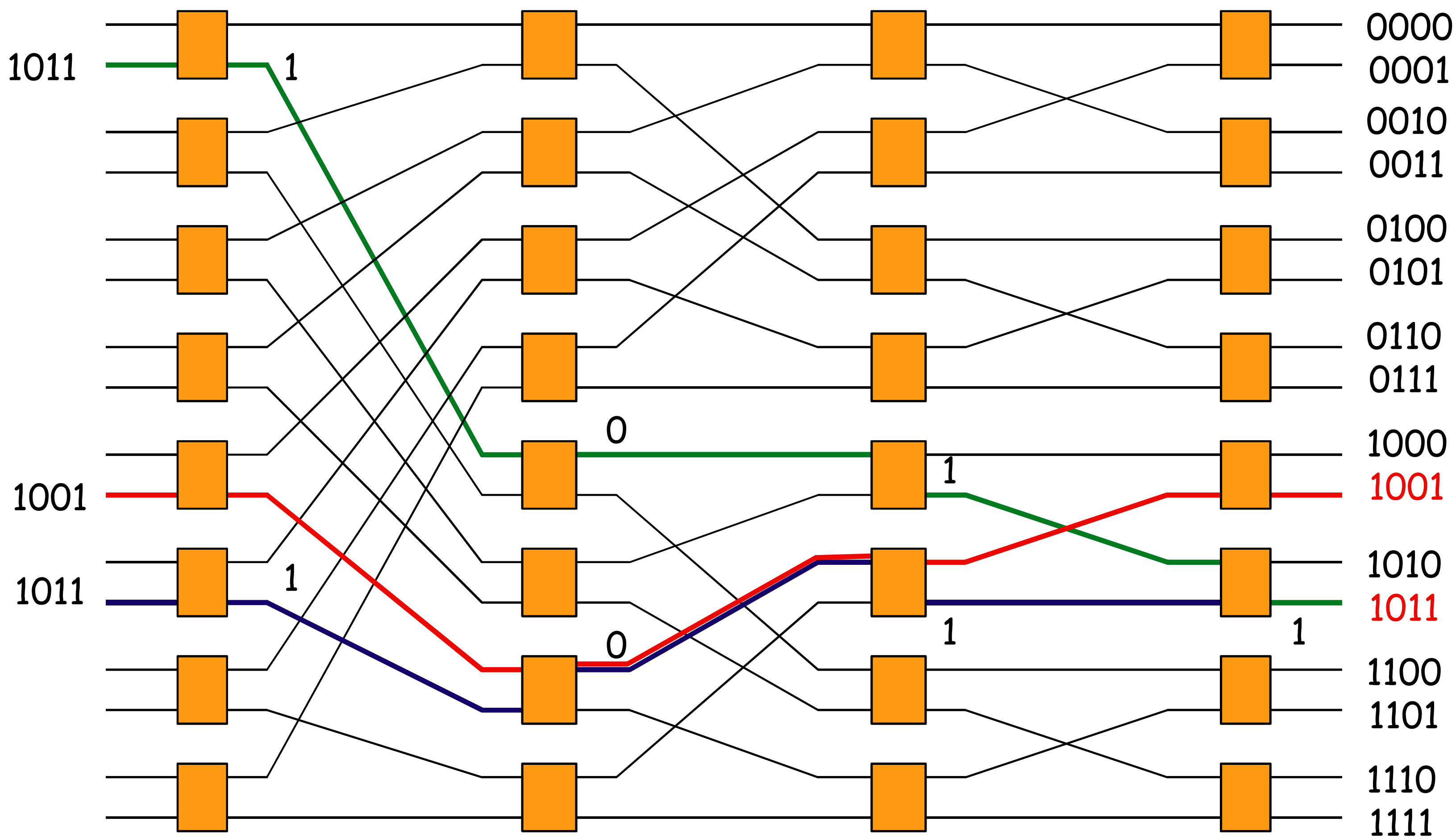


- Internally & externally non-blocking: "m" multiple paths between I/O pair.
- Each output port consists of a concentrator and a shared buffer.
- Advantage: high performance, multicast-ready
- Disadvantage: costly to implement, switch elements $O(n^2)$, difficult to scale

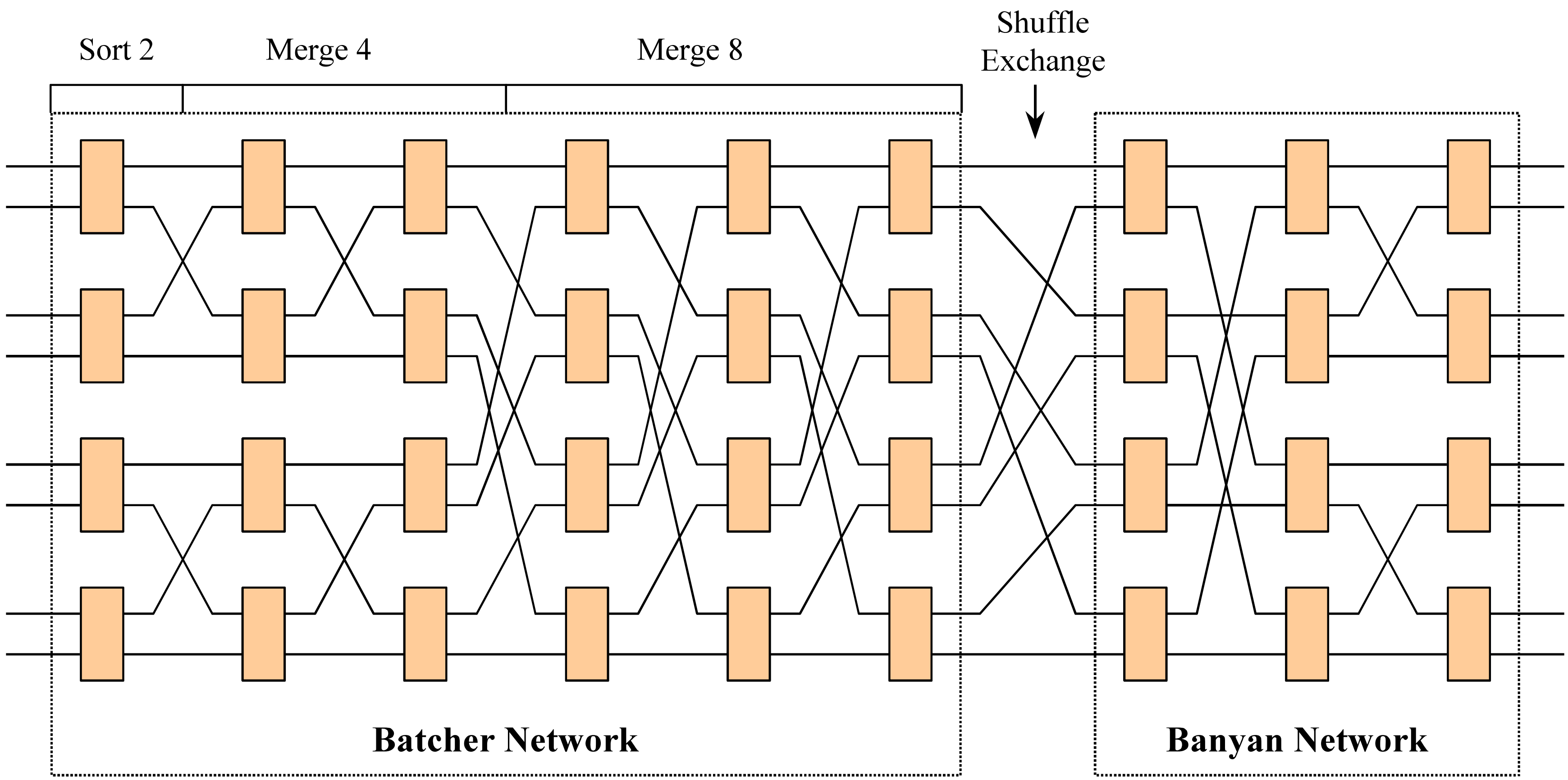
Clos Network



Delta Network Switching System



Batcher-Banyan Switching System



Batcher-Banyan Switching System

- Category:
  - » Space switching
  - » Internally non-blocking Multiple Interconnection Network (MIN)
  - » No internal buffering required
  - » Output contention still a possibility, can be solved by using
    - additional arbitration logic
    - input buffering
    - output buffering with fabric speed up.

Batcher-Banyan Switching System

- Batcher Network = sorting network
  - » Sorts all cells according to destination
  - » Cells to same destination are adjacent.
- Banyan Network = self-routing network
  - » No internal blocking, if cells are sorted at inlet
  - » No external blocking, if only one cell for each outlet.
- Output port contention solved using 3 phase algorithm
  - » Arbitration phase
  - » Acknowledgement phase
  - » Sending phase

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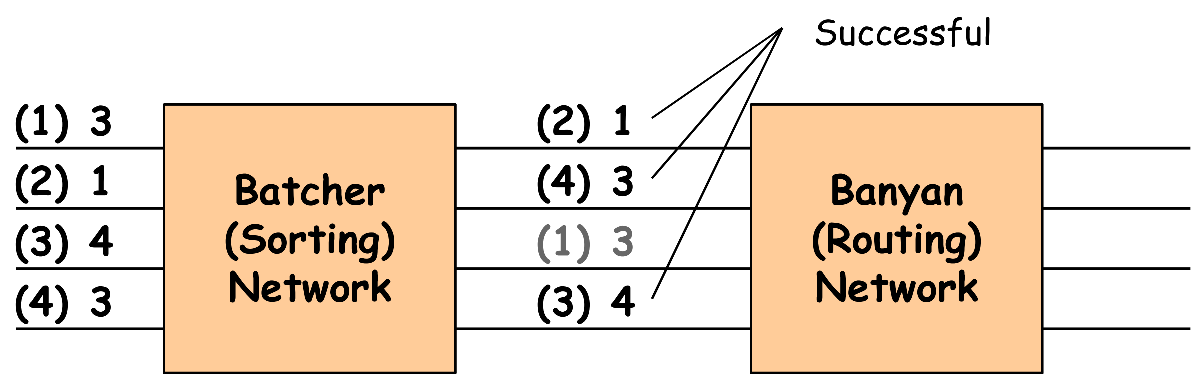
Batcher-Banyan Switching System



- Phase I: Send and Resolve Request
  - » Send source-destination pair through sorting network
  - » Sort destination in non-decreasing order
  - » Purge adjacent requests with same destination.

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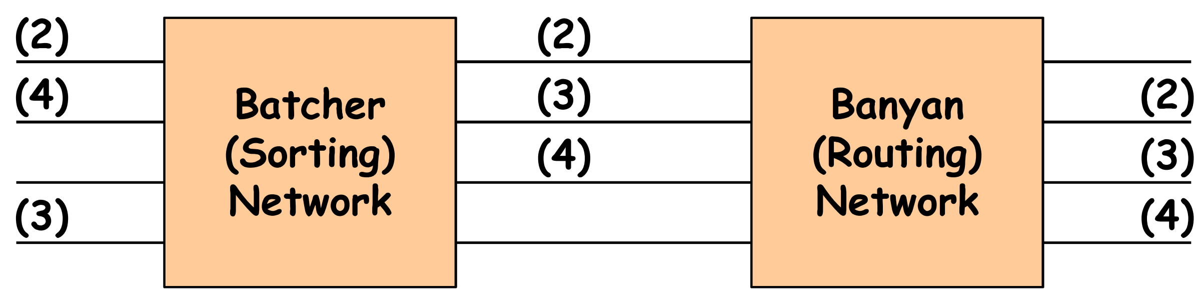
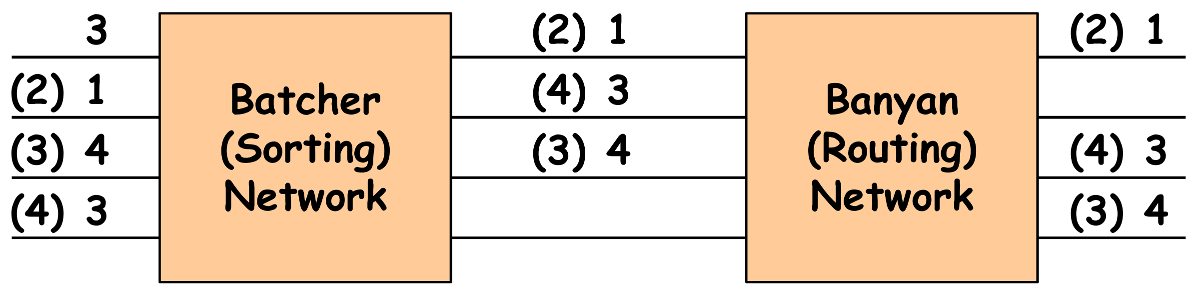
Batcher-Banyan Switching System



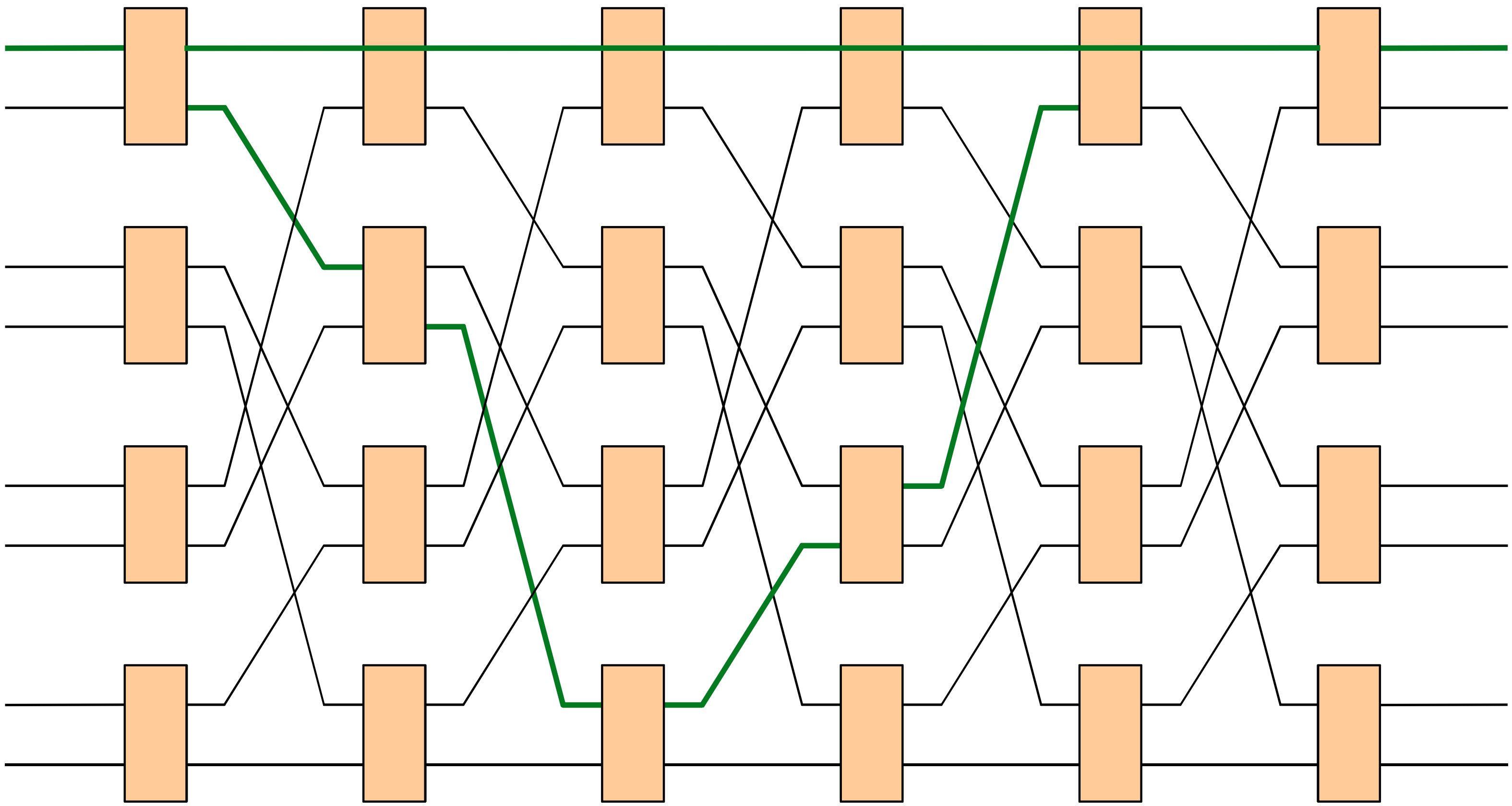
- Phase II: Acknowledge winning port
  - » Send ACK with destination to the port with winning contention
  - » Route ACK through Batcher-Banyan network

Batcher-Banyan Switching System



- Phase III: Send Packet
  - Acknowledged ports send packet through Batcher Banyan network
  - Buffers at port controller

Augmented Banyan



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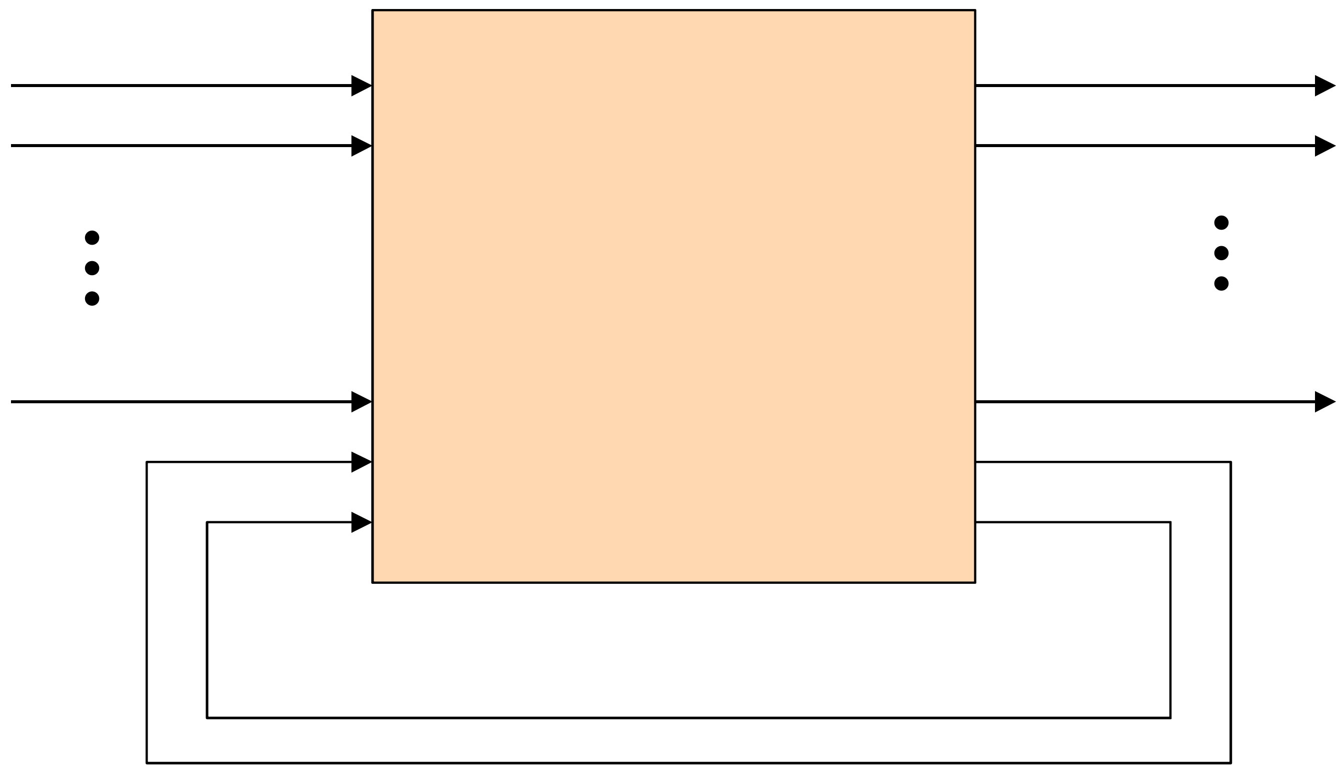
Recirculation Network



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Blocking Effects in Switches

- *Blocking* is a fundamental problem of switching
  - » occurs when two or more cells contend for the same resources
    - paths through the switching fabric (internal blocking)
    - output ports (external blocking / output contention)
- *Queueing* is the solution to the blocking effect
  - » cells are temporarily stored in buffer memory until it can be safely transported to the intended output port.
  - » basic queueing techniques
    - input, output, internal, shared buffering.

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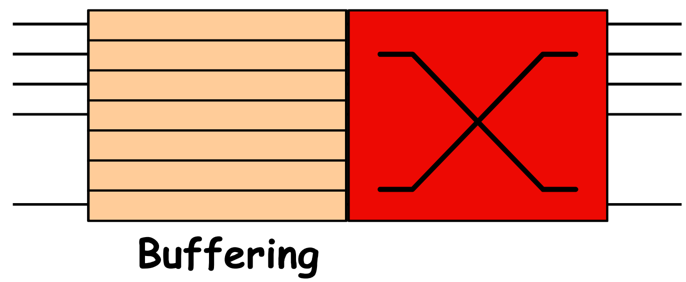
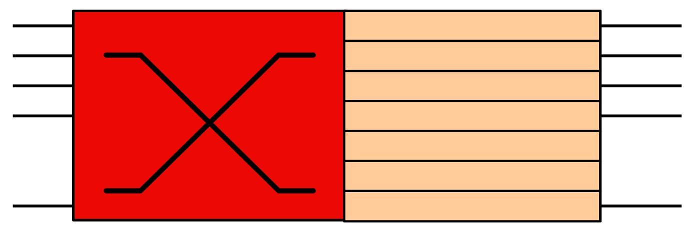
Input Buffering



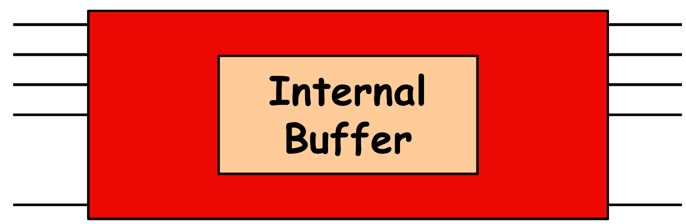
- All arriving cells are queued at the input
  - » cells are queued until the switch indicates that the cell has been successfully switched.
  - » cells that fail to reach the output port stay in the buffer and can be tried again on the next round of arbitration.
- Advantage: Simple to implement.
- Disadvantage: Head of Line (HOL) blocking.

Output Buffering



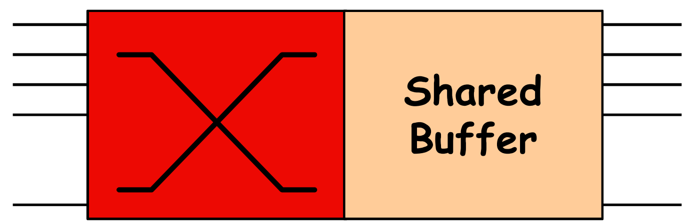
- Assumption: All arriving cells must be switched to the output for buffering
  - » When two or more cells are switched to the same output port in a cycle, extra cells are buffered until it can be transmitted.
  - » Switching transfer speed must be performed at N times the inlet speed. If transfer speed < N * inlet speed, then internal cell loss will occur.
- Advantage: Avoids HOL blocking, no arbitration required
- Disadvantage: Switch fabric may be difficult to implement

Internal Buffering



- Buffering of cells is done within the internal fabric of the switch.
- Advantage: Hides the concept of buffering
- Disadvantage: Difficult to implement efficiently.
- This technique is hardly used.

Shared Buffering



- Assumption: All arriving cells are switched to the shared output buffer.
- When two or more cells are switched to the same output port in a cycle, extra cells are buffered until it can be transmitted.
- Advantage: Memory requirement is greatly reduced, no HOL blocking
- Disadvantage: Limited memory bandwidth.